Submitted to the Annals of the Department of Statistics and Operations Research

ESTIMATING EXPECTED GOALS: A COMPARISON OF LOGISTIC REGRESSION AND RANDOM FOREST

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Expected goals (xG) have become a central metric in modern football analytics, providing a probabilistic assessment of shot quality and chance conversion. This thesis investigates how well different modelling approaches can estimate xG using publicly available event data. The analysis is based on StatsBomb Open Data for all matches of the FIFA World Cups 2018 and 2022. Shot events from 2018 are used to train the models and shots from 2022 serve as an independent test set in a temporal validation framework.

Two model classes are considered: a logistic regression model as an interpretable statistical baseline and a random forest classifier as a flexible machine learning method. Both models use geometric shot features and contextual variables, they are evaluated against StatsBomb's proprietary xG estimate as an external benchmark. Predictive performance is assessed using the Area Under the ROC Curve (AUC), the Brier score and log loss, complemented by classification metrics at the Youden-optimal threshold.

Empirically, the logistic regression model closely approaches the performance of the StatsBomb xG benchmark, while the random forest performs noticeably worse in terms of both discrimination and calibration. These results suggest that, with World Cup shot data and a carefully chosen feature set, a relatively simple and transparent logistic regression model can provide competitive xG estimates without the need for highly complex machine learning architectures.

1. Introduction. Football is an ever-evolving sport. Sometimes, fundamental changes occur suddenly, for example through rule changes. At other times, change is driven by influential coaches whose ideas are widely copied. Pep Guardiola's teams, for instance, have reshaped positional play and possession structures at the highest level, inspiring tactical adaptations across leagues and continents. These developments are not only grounded in intuition and experience, they increasingly result from systematic analysis and data driven insights.

Over the last decades, the use of statistics and data in football has grown substantially, see (Mead, O'Hare and McMenemy, 2023; Manley et al., 2025). Clubs routinely employ data analysts and scouting processes rarely rely on visual impressions alone. Quantitative metrics are used to evaluate how well a player fits a team's tactical approach, to inform recruitment of profiles to look for and match preparation. The desire to gain even small competitive advantages has led to a proliferation of metrics describing different aspects of the game.

This development is particularly palpable at the highest level of the sport. The FIFA World Cup is the most prestigious and most-watched football tournament globally, it concentrates the world's best players and nations in a short, high-stakes competition. Performances at World Cups have a disproportionate impact on reputations and transfer values, which makes understanding data especially relevant.

Among the many performance indicators used in modern football analytics, expected goals (xG) has become one of the most prominent. Intuitively, xG seeks to answer the question, how likely it is to score a goal from a given shot situation. Formally, each shot is assigned a probability between zero and one that represents the estimated chance of scoring from that situation (cf. Rathke, 2017). Aggregated over many shots, xG can be used to evaluate the quality of chances created and conceded by teams or to assess whether individual players are finishing above or below expectation. For strikers, the number of goals scored relative to expected goals is increasingly used as a measure of finishing efficiency. For goalkeepers, the comparison between expected and actual goals conceded can serve as an indicator of shot-stopping performance.

In the academic and applied analytics literature, a variety of xG models have been proposed. Early approaches relied primarily on logistic regression with basic geometric features such as distance and angle to goal (e.g. Rathke, 2017). More recent work has incorporated richer contextual information and has explored a range of machine learning techniques such as gradient boosting and random forests, see (Mead, O'Hare and McMenemy, 2023). Despite these advances, there is still an active debate about the trade-off between model complexity, interpretability and the quality of probabilistic predictions.

At the same time, most proprietary xG models used by clubs and data providers are not publicly documented and their underlying data and features are not accessible to external researchers. Open data initiatives such as StatsBomb Open Data play a crucial role in enabling reproducible research and allowing academics and independent analysts to work with xG models on a common footing.

This thesis investigates the modelling of expected goals using event-level shot data from the FIFA World Cups 2018 and 2022, provided by StatsBomb Open Data (cf. StatsBomb, 2019). Two approaches are compared: logistic regression as a classical statistical baseline and random forest as an established machine learning method. Both models are trained on shots from the 2018 World Cup and evaluated on shots from the 2022 World Cup in a temporal validation framework. Model performance is assessed using the Area Under the ROC Curve (AUC), the Brier score and log loss, and both models are benchmarked against the internal StatsBomb xG estimate.

The remainder of this paper is structured as follows. Section 2 describes the data, variable selection and preprocessing steps. Section 3 introduces the modelling framework, the validation strategy and the evaluation metrics. Section 4 presents the empirical results. Section 5 discusses the findings, outlines practical implications and limitations, and suggests directions for future work, with particular emphasis on the relevance of xG modelling in a World Cup context.

2. Data.

2.1. *Data Source and Dataset.* The data used in this thesis were obtained from StatsBomb Open Data (cf. StatsBomb, 2019), a publicly available repository, providing detailed event-level data for football matches. In StatsBomb's event model, each on-ball action (for example a shot or tackle) is recorded as a separate entry

with spatial coordinates and contextual metadata (cf. StatsBomb, 2019). A shot is defined as any deliberate attempt to score a goal. All data were accessed via the `StatsBombR` package in R (cf. R Core Team, 2022).

The dataset comprises all matches from the FIFA World Cup 2018 (Russia) and the FIFA World Cup 2022 (Qatar), totalling 3,200 recorded shots. Two filtering steps were applied prior to analysis. First, penalty kicks were excluded, as they represent a fundamentally different shooting situation with a fixed distance and no defensive pressure, reducing the dataset to 3,068 shots. Second, one additional observation was removed due to missing goalkeeper freeze frame data required to compute the `AngleDeviation` variable, resulting in a final dataset of 3,067 shots. Geometric features such as distance to goal (`DistToGoal`), angle to goal (`AngleToGoal`), and goalkeeper angle deviation (`AngleDeviation`) were computed automatically by the `allclean()` function of the `StatsBombR` package (cf. StatsBomb, 2019).

2.2. *Variables.* The final model dataset contains 3,067 shots and includes the following variables. Table 1 provides an overview of all features used in the analysis; full variable specifications follow StatsBomb's event definitions (cf. StatsBomb, 2019).

TABLE 1
Overview of model variables.

Variable	Type	Description
`DistToGoal`	Numeric	Distance from shooter to centre of goal
`AngleToGoal`	Numeric	Opening angle between shooter and goalposts
`AngleDeviation`	Numeric	Angular displacement of the goalkeeper from the shooter's direct line to goal centre
`InCone.GK`	Integer	Number of players on the shooter–GK line
`DefendersInCone`	Integer	Opposing outfield players in the shot triangle
`shot.body_part`	Categorical	Body part used (foot, head, etc.)
`shot.technique`	Categorical	Technique (normal, volley, half volley, etc.)
`shot.type`	Categorical	Situation (open play, free kick, corner)
`under_pressure`	Boolean	Shooter under defensive pressure
`shot.first_time`	Boolean	First-time shot without ball control
`shot.one_on_one`	Boolean	One-on-one situation with goalkeeper

2.3. *Target variable.* The binary target variable `goal` indicates whether a given shot resulted in a goal (`goal = 1`) or not (`goal = 0`). In the final dataset of 3,067 shots, 287 shots were scored and 2,780 were not scored, corresponding to a conversion rate of approximately 9.35%. This strong class imbalance is typical for shot data in football, see (Mead, O'Hare and McMenemy, 2023) and has important implications for model evaluation, as naive classifiers that always predict "no goal" would achieve an accuracy of 90.65% on this dataset while correctly identifying zero goals, demonstrating that accuracy alone is a misleading measure of model quality under class imbalance.

TABLE 2
Distribution of the target variable `goal`.

Outcome	Count	Proportion
No goal (0)	2,780	90.65%
Goal (1)	287	9.35%
Total	3,067	100%

TABLE 3
Distribution of shots by body part (World Cups 2018 and 2022).

Body part	Count	Proportion
Right foot	1,557	50.77%
Left foot	921	30.03%
Head	582	18.32%
Other	27	0.88%
Total	3,067	100%

Most shots originate from open play (95.76%), with a smaller share from free kicks(4.18%). Corner shots are extremely rare in the final dataset (0.06%), reflecting the fact that no corner shots are present in the 2018 training data.

2.4. *Data Preparation.* Prior to model fitting, three boolean variables required additional preprocessing. The variables `under_pressure`, `shot.first_time`, and `shot.one_on_one` contained missing values (`NA`) in cases where the respective condition was not recorded by StatsBomb. Following standard practice for boolean event indicators, these missing values were replaced by `FALSE`, as the absence of a recorded event implies that the condition did not apply (cf. StatsBomb, 2019). Additionally, one observation was removed due to missing goalkeeper freeze frame data required to compute `AngleDeviation`, resulting in a final dataset of 3,067 shots.

Geometric variables (`DistToGoal`, `AngleToGoal`, `AngleDeviation`) were computed by the `allclean()` function using the exact shot and goalkeeper coordinates from the StatsBomb freeze frame data (cf. StatsBomb, 2019). The variables `DistToKeeper`, `AngleToKeeper`, `density`, and `density.incone` were excluded due to high pairwise correlations ($r > 0.7$) with retained variables, which would introduce multicollinearity into the model, see (Kim, 2019). The StatsBomb internal xG estimate (`shot.statsbomb_xg`) is retained in the dataset solely as a benchmark for model evaluation and is not used as a predictor.

2.5. *Descriptive statistics.* Table 4 summarises basic descriptive statistics for the key numeric variables in the final dataset. Distances and angles show a wide range of values, indicating that shots are taken from a diverse set of positions and orientations. The low mean values of the geometric variables reflect that most shots are taken from outside the immediate vicinity of the goal.

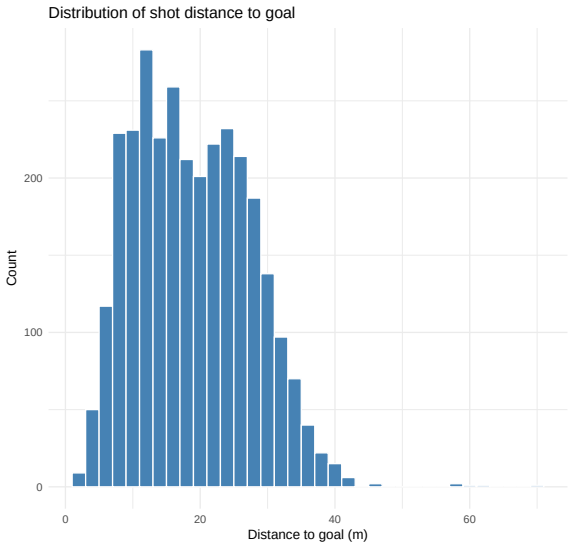


FIG 1. *Distribution of shot distance to goal across both tournaments.*

TABLE 4
Descriptive statistics of selected numeric variables (World Cups 2018 and 2022).

Variable	Min	1st Qu.	Median	Mean	Max
DistToGoal	1.32	12.04	18.28	18.92	70.83
AngleToGoal	1.17	62.17	90.00	90.15	180.00
AngleDeviation	0.00	6.34	15.95	24.03	167.20
InCone.GK	0.00	1.00	1.00	1.45	4.00
DefendersInCone	0.00	0.00	1.00	1.06	7.00

Figure 1 shows that most shots are taken from distances between 10 and 30 metres, with relatively few attempts from very close or very long range. This pattern is consistent with typical shot distributions in elite football.

3. Methodology.

3.1. *Model framework.* The objective of this study is to estimate, for each shot, the conditional probability that it results in a goal given a set of observable shot characteristics. Since `goal` is a binary variable taking values in $\{0, 1\}$, its conditional expectation reduces to

$$\begin{aligned}\mathbb{E}[\text{goal} \mid \text{features}] &= 0 \cdot P(\text{goal} = 0 \mid \text{features}) + 1 \cdot P(\text{goal} = 1 \mid \text{features}) \\ &= P(\text{goal} = 1 \mid \text{features}),\end{aligned}$$

which directly defines the expected goals (xG) value for the corresponding shot.

Two classification models are employed and compared: a logistic regression model and a random forest classifier. Logistic regression serves as a baseline due to its

prevalence in the xG literature and its high interpretability, as regression coefficients can be related directly to changes in log-odds and odds ratios, see (Rathke, 2017; Mead, O'Hare and McMenemy, 2023). The random forest is included as a more flexible, non-parametric alternative that can approximate non-linear decision boundaries and capture interaction effects between features without the need to specify them manually, see (Mead, O'Hare and McMenemy, 2023).

3.2. *Temporal validation.* To evaluate out-of-sample predictive performance, a temporal validation strategy is adopted. All shots from the FIFA World Cup 2018 are used as the training set, while all shots from the FIFA World Cup 2022 form a held-out test set. Unlike a random train–test split, this setup mimics a realistic forecasting scenario in which a model is trained on past tournaments and evaluated on a later tournament, thereby providing a more stringent assessment of generalisation performance, see (Mead, O'Hare and McMenemy, 2023).

The season indicator is only used to define this temporal split and is excluded from the subsequent model estimation. No information from the 2022 test set is used for model training or hyperparameter tuning.

3.3. *Model training.* Both models are trained exclusively on the World Cup 2018 data. For the logistic regression model, goal is used as a binary response variable and all selected predictors (geometric variables, categorical shot descriptors and boolean indicators) enter the model as covariates. The model is estimated by maximum likelihood under a binomial logit link. Categorical predictors (`shot.body_part`, `shot.technique`, `shot.type`) are included via R's formula interface, which internally creates dummy variables with one reference category per factor. Model fit on the training set is summarised using McFadden's pseudo-R^2, and multicollinearity is assessed using Generalised Variance Inflation Factors (GVIF). The StatsBomb xG variable (`shot.statsbomb_xg`) is never used as a predictor and serves only as an external benchmark in the evaluation phase.

In terms of complexity, the logistic regression includes a modest number of parameters relative to the sample size (around 1,600–1,700 shots in the 2018 training set), which helps to mitigate overfitting. Each numeric predictor enters the model linearly, while categorical variables contribute several dummy coefficients, leading to a total of roughly 15–20 regression parameters.

For the random forest model, the target is treated as a two-class factor (`No`, `Goal`). Hyperparameter tuning is carried out using five-fold cross-validation on the 2018 training data, implemented with the `caret` package in R. The only tuned hyperparameter is the number of randomly selected predictors considered at each split (mtry). Other settings such as minimum node size and maximum tree depth are left at their defaults. With only around 1,600 training observations, an extensive multi-parameter search would risk overfitting to the cross-validation folds and mtry is generally considered the most impactful hyperparameter for random forest performance (cf. Breiman, 2001). A small grid $\{2, 3, 4\}$ is evaluated, and for each value of mtry the cross-validated log loss (multinomial negative log-likelihood) of the predicted class probabilities is computed. The configuration with the smallest cross-validated log loss (here mtry $= 4$) is selected as the final random forest model.

The number of trees is fixed at 500, following common practice in the literature, see, for instance, (Breiman, 2001).

3.4. *Evaluation metrics.* After model training and hyperparameter tuning on the 2018 data, the final logistic regression and random forest models are evaluated on the held-out 2022 test set. Predictive performance is assessed using three complementary metrics, see (Mead, O'Hare and McMenemy, 2023).

The Area Under the Receiver Operating Characteristic Curve (AUC-ROC) measures the ability of a model to discriminate between goals and non-goals across all possible classification thresholds. Formally,

$$\text{AUC} = P(\hat{p}_i > \hat{p}_j \mid y_i = 1,\, y_j = 0),$$

where $\hat{p}_i$ and $\hat{p}_j$ are the predicted probabilities for a randomly chosen goal and non-goal, respectively. This equals the area under the ROC curve, see (Fawcett, 2006).

The Brier score quantifies the mean squared difference between predicted probabilities and observed outcomes:

$$\text{BS} = \frac{1}{n}\sum_{i=1}^{n}(\hat{p}_i - y_i)^2,$$

where $y_i \in \{0, 1\}$ is the observed outcome and $\hat{p}_i$ the corresponding predicted probability. Lower Brier scores indicate that predicted probabilities are, on average, closer to the actual binary outcomes.

The log loss (binary cross-entropy) is defined as

$$\text{LL} = -\frac{1}{n}\sum_{i=1}^{n}\big[y_i \log(\hat{p}_i) + (1 - y_i)\log(1 - \hat{p}_i)\big].$$

Log loss penalises incorrect and overconfident predictions particularly strongly: when the model assigns a high probability to a non-goal ($y_i = 0$, $\hat{p}_i \approx 1$), the term $\log(1 - \hat{p}_i) \to -\infty$, resulting in a very large penalty. In this study, log loss is used as the tuning metric in the cross-validation procedure for the random forest and as an additional performance measure on the test set.

In addition, for interpretability in a classification setting, accuracy, sensitivity and specificity are reported at the threshold that maximises Youden's index, i.e. the sum of sensitivity and specificity minus one, derived from the ROC curve. Finally, both models are benchmarked against the internal StatsBomb xG estimate (`shot.statsbomb_xg`) on the same test set to contextualise performance relative to a state-of-the-art industry model.

4. Results.

4.1. *Descriptive performance of the models.* Table 5 reports the out-of-sample performance of the three xG models on the FIFA World Cup 2022 test set: the

bespoke logistic regression model, the random forest classifier, and the StatsBomb internal xG estimate used as an external benchmark. Model performance is measured in terms of the Area Under the ROC Curve (AUC), the Brier score and, for later discussion, classification metrics at the optimal Youden threshold.

TABLE 5
Out-of-sample performance on World Cup 2022 (shot-level).

Model	AUC-ROC	Brier score	Comment
StatsBomb xG	0.8145	0.0763	External industry benchmark
Logistic regression	0.7960	0.0812	Own interpretable xG model
Random forest (CV, mtry = 4)	0.7323	0.0866	Non-linear model, log-loss tuned

The StatsBomb xG benchmark attains the highest AUC (0.8145) and the lowest Brier score (0.0763), indicating both strong discriminative ability and good overall calibration. The custom logistic regression model comes remarkably close, achieving an AUC of 0.7960 and a Brier score of 0.0812. In contrast, the random forest exhibits substantially lower discriminative performance (AUC = 0.7323) and a higher Brier score (0.0866), even though it has been tuned via cross-validated log loss.

4.2. *Logistic regression performance.* The logistic regression model demonstrates a solid fit on the training data from World Cup 2018. McFadden's pseudo-R^2 is approximately 0.166, which is a typical magnitude for logistic models in sports applications (cf. McFadden, 1974) and indicates a substantial improvement over the intercept-only model. Generalised Variance Inflation Factors (GVIF) are close to one for all predictors (all $\text{GVIF}^{1/(2\cdot\text{df})} < 1.3$), suggesting that multicollinearity is not a concern for this specification.

Out-of-sample, the logistic regression achieves an AUC of 0.7960 on the 2022 test set, indicating good discrimination between shots that result in goals and those that do not. The Brier score of 0.0812 shows that the predicted probabilities are reasonably well calibrated, though slightly less so than the StatsBomb xG benchmark (Brier score 0.0763). The distribution of predicted probabilities is highly concentrated at low values, with a median around 0.054 and a maximum of approximately 0.77, which aligns with the low overall conversion rate of shots at this level.

Using the Youden index (cf. Youden, 1950) to determine an optimal classification threshold yields a cut-off of roughly 0.081. At this threshold, the logistic regression produces 866 true negatives, 410 false positives, 37 false negatives and 115 true positives on the test set, corresponding to an overall accuracy of about 68.7%. The model thus identifies a relatively large share of actual goals (high sensitivity) but at the cost of classifying many low-probability shots as potential goals, which is consistent with the conservative threshold induced by the strongly imbalanced class distribution.

4.3. *Random forest performance.* The random forest model is tuned on the 2018 training data using five-fold cross-validation and log loss as the optimisation criterion. Table 6 summarises the cross-validated log loss for the three candidate values of the mtry parameter.

TABLE 6
Cross-validated log loss for random forest on World Cup 2018 (five-fold CV).

mtry	Log loss (CV)
2	0.4177
3	0.3416
4	0.2793

The log loss decreases monotonically as mtry increases from two to four, with the lowest value obtained at mtry = 4. Consequently, this configuration is selected as the final model and is re-fitted on the full 2018 training set. Despite this careful tuning, the random forest achieves an AUC of only 0.7323 and a Brier score of 0.0866 on the 2022 test set, substantially worse than both the logistic regression and the StatsBomb benchmark.

At the Youden-optimal threshold of approximately 0.081, the random forest attains an accuracy of about 77.5%, a sensitivity of 57.9% and a specificity of 79.8% on the test set. Compared to the logistic regression, the random forest yields higher overall accuracy and specificity but lower sensitivity, meaning that it is more conservative in predicting goals and misses a larger fraction of actual goals. This asymmetry is noteworthy given that, from a tactical or coaching perspective, detecting goal-scoring opportunities (sensitivity) is often more valuable than correctly rejecting non-threatening shots.

TABLE 7
Classification metrics at the Youden-optimal threshold (World Cup 2022 test set).

Model	Accuracy	Sensitivity	Specificity
Logistic regression	0.69	0.76	0.68
Random forest	0.775	0.579	0.798

Table 7 summarises accuracy, sensitivity and specificity at the Youden-optimal threshold for both models. The random forest achieves higher overall accuracy and specificity, but at the cost of substantially lower sensitivity. In contrast, the logistic regression detects a larger proportion of actual goals, which may be preferable in applications where identifying goal-scoring opportunities is more important than minimising false positives.

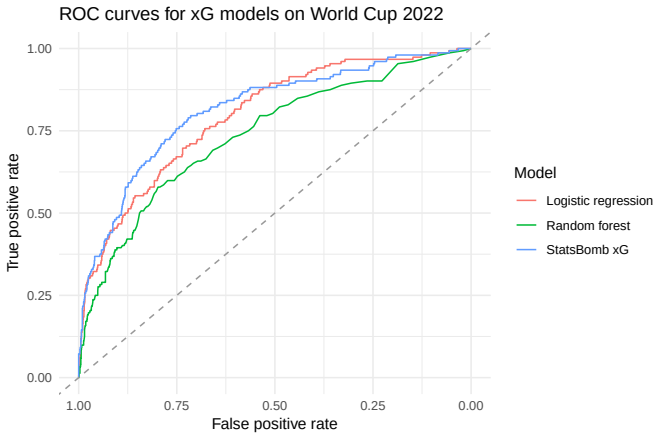


FIG 2. *ROC curves for the three xG models on the World Cup 2022 test set.*

Figure 2 visualises the ROC curves of the three models on the 2022 test set. The StatsBomb xG curve consistently lies above the other two, followed closely by the logistic regression, while the random forest curve is clearly closer to the diagonal, reflecting its lower AUC values.

4.4. *Calibration of the logistic regression model.* Figure 3 shows a calibration plot for the logistic regression model on the 2022 test set. Shots are grouped into bins of similar predicted xG values, for each bin the average predicted probability is compared to the observed goal frequency. The points lie reasonably close to the diagonal, indicating that the predicted probabilities are broadly consistent with the observed goal frequencies across the range of xG values, although there is some deviation in the highest-probability bins. The concentration of points towards the lower end of the x-axis reflects the skewed distribution of predicted probabilities: with a median predicted xG of around 0.054, most quantile bins fall in the low-probability range.

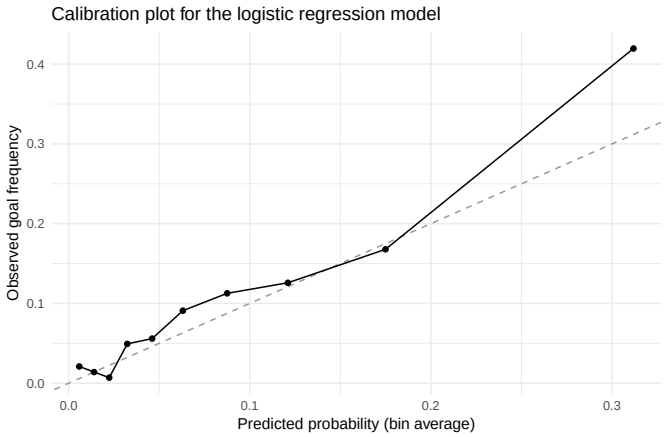


FIG 3. *Calibration plot for the logistic regression model on the World Cup 2022 test set. Each point represents a bin of shots with similar predicted probabilities; the dashed line indicates perfect calibration.*

4.5. *Comparison with StatsBomb xG.* The StatsBomb internal xG estimate serves as a strong reference point, as it is produced by a proprietary model trained on a substantially larger and more diverse dataset than the two models developed in this study. On the World Cup 2022 test set, StatsBomb xG achieves the best combination of discrimination (AUC = 0.8145) and calibration (Brier score = 0.0763).

Nevertheless, the custom logistic regression model narrows the gap considerably, reaching an AUC only about 0.02 points lower and a Brier score only about 0.005 points higher than the StatsBomb benchmark. This suggests that, given well-chosen geometric and contextual features, a relatively simple and interpretable model can approximate the performance of a state-of-the-art industry xG model on a large international tournament. In contrast, the random forest does not provide an advantage in this setting; its lower AUC and higher Brier score indicate that the additional flexibility does not translate into better generalisation performance on the 2022 World Cup data.

5. Discussion.

5.1. *Interpretation of model performance.* The empirical results show a clear ordering of model performance on the World Cup 2022 test set: the proprietary StatsBomb xG model performs best, followed closely by the custom logistic regression model, while the cross-validated random forest trails behind both in terms of discrimination and calibration. This pattern is consistent across AUC and Brier score and is therefore unlikely to be an artefact of any single metric.

From a practical perspective, the proximity of the logistic regression model to the StatsBomb benchmark is particularly noteworthy. Despite being estimated on a much smaller and less diverse dataset, the logistic model achieves AUC that is only about 0.02 points lower and a Brier score that is only marginally higher than those

of the industry model. This suggests that a carefully specified, interpretable model can capture a substational portion of the information relevant for shot quality, even without the full data volume available to a commercial provider.

In contrast, the random forest model does not deliver the expected performance gains over the logistic regression. Although it is more flexible and has been tuned via cross-validated log loss, its AUC is notably lower and its Brier Score higher than those of the logistic model. This indicates that, in the present setting, additional model flexibility does not translate into better generalisation to future tournaments.

5.2. *Why the random forest underperforms.* There are severeal plausible reasons why the random forest underperforms relative to the logistic regression in this application. First, the effective sample size is moderate: the training set from the 2018 World Cup comprises only around 1600 to 1700 shots. While this is sufficient for estimating a logistic model with a dozen predictiors, it offers less room for a high-variance ensemble method to learn complex non-linearities without overfitting.

Second, the chosen feature set may simply not exhibit strong non-linear patterns that would favour a tree-based method. The core predictors are essentially geometric (distance, angle, deviation to goalkeeper) and simple positional or contextual indicators. For these variables, the relationship with scoring probability is often monotone and smooth,because of that logistic regression can approximate well with linear terms and a logit link. If most of the signal is captured by such low-dimensional, roughly monotonic effects, the marginal benefit of additional non-linearity is limited.

Third, the random forest used here is relatively simple in terms of hyperparameter space. Only the mtry parameter is tuned over a small grid, while other potentially influential settings (such as node size, maximum tree depth or class weights) remain at their defaults. A more extensive tuning procedure or alternative tree-based methods (e.g. gradient boosting machines) might extract more performance, but would also increase complexity and the risk of overfitting further on such a small dataset.

Taken together, the arguments suggest that, for the given data and feature set, the logistic regression strikes a favourable balance between bias and variance, see (Breiman, 2001). The random forest, while more flexible in principle, does not find enough additional structure in the data to justify its complexity and therefore yield weaker out-of-sample performance,

5.3. *Implications for xG modelling with open data.* The fact that a relatively simple logistic regression model can approximate the performance of a proprietary StatsBomb xG model has several implications for practitioners and researchers working with open football data. First, it supports the view that meaningful and practically useful xG models can be built using only publicly available event data and standard statistical techniques. This lowers the barrier to entry for those who do not have access to large proprietary datasets.

Second, the interpretability of the logistic model is a non-trivial advantage. Because the coefficients can be examined directly, it is possible to quantify the marginal effects of distance, angle, defensive pressure or one-on-one situations on

scoring probability. These insights can be communicated to others in a more transparent way than the predictions of a black-box model, even if the latter achieves slightly better aggregate performance.

Third, the underperformance of the random forest serves as a cautionary note against assuming that more complex models automatically lead to better results. In domains with limited data and relatively low signal-to-noise ratios, simpler models with well-understood behaviour may offer more robust and reliable predictions than highly flexible machine learning algorithms.

5.4. *Limitations and possible extensions.* While the present modelling framework is deliberately simple and transparent, it also has important limitations. The analysis is restricted to two World Cup tournaments, which are elite competitions with specific tactical and contextual characteristics. It is therefore not guaranteed that the models generalise to lower levels of football. Extending the dataset to include multiple seasons and competitions would allow for more robust estimation and validation.

Moreover, the feature set is limited to variables derived from event data and basic geometric constructs. Modern xG models often include richer information, such as defender and goalkeeper positions from tracking data (cf. Mead, O'Hare and McMenemy, 2023), shot speed, body orientation, or preceding actions. Incorporating such features could further narrow the gap to the StatsBomb benchmark or even surpass it, but would require access to more detailed data and more sophisticated feature engineering.

Finally, only two model classes have been considered: logistic regression and random forest. Other approaches, such as gradient boosting, calibration methods (e.g. isotonic regression or Platt scaling; see (Niculescu-Mizil and Caruana, 2005)) or neural networks, could be explored in future work. In particular, combining a flexible base learner with explicit calibration techniques might improve probabilistic accuracy while retaining good discrimination. However, given the constrained data setting in this study, the present results already illustrate that relatively simple, interpretable models can provide competitive xG estimates in an international-tournament context.

6. Conclusion. This thesis set out to investigate how well different modelling approaches can estimate expected goals (xG) using publicly available event data. Using StatsBomb Open Data from the FIFA World Cups 2018 and 2022, a logistic regression model and a random forest classifier were trained on the 2018 tournament and evaluated on the 2022 tournament within a temporal validation framework. Both models were benchmarked against StatsBomb's internal xG estimate, which serves as an industry reference.

The empirical results show a clear hierarchy of performance. The proprietary StatsBomb xG model attains the highest AUC and the lowest Brier score on the 2022 test set, indicating strong discrimination between goals and non-goals and good overall calibration. The custom logistic regression model comes remarkably close, with only slightly lower AUC and slightly higher Brier score. In contrast, the

cross-validated random forest performs noticeably worse in both metrics, despite its greater flexibility and the use of log-loss-based hyperparameter tuning.

These findings suggest that, for World Cup shot data and a carefully chosen set of geometric and contextual features, a relatively simple and interpretable logistic regression model can provide competitive xG estimates without requiring highly complex machine learning architectures. This has practical implications for clubs and analysts working primarily with open event data: meaningful xG models can be built using standard statistical tools, even in the absence of large proprietary datasets or tracking information.

At the same time, the study has several limitations. The analysis is restricted to two international tournaments and to features derived from event data; it does not incorporate richer positional information or more advanced model classes such as gradient boosting or neural networks. Future work could extend the dataset to multiple seasons and leagues, explore additional feature engineering using tracking data, and investigate more flexible models in combination with explicit calibration techniques, see (Niculescu-Mizil and Caruana, 2005). Nonetheless, the present results already demonstrate that transparent, data-driven models can capture much of the variation in shot outcomes at the highest level of football.

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